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clc;
clear;
close all;

% Parameters
Am = 1;
fm = 5;
Ac = 5;
fc = 50;
fs = 20 * fc;

% Time
t = 0:1/fs:4/fm;

% Message signal
m = Am * sin(2*pi*fm*t);

% Carrier signal
c = Ac * sin(2*pi*fc*t);

% Modulation index
mu = Am/Ac;

% AM signal
am = Ac * (1 + mu*sin(2*pi*fm*t)) .* sin(2*pi*fc*t);

% FFT
N = length(am);
f = (-N/2:N/2-1) * fs/N;
AM_FFT = abs(fftshift(fft(am))) / N;

% USB and LSB
USB = fc + fm;
LSB = fc - fm;

% Plot AM frequency spectrum
figure('Name', 'AM Frequency Spectrum', 'NumberTitle', 'off');

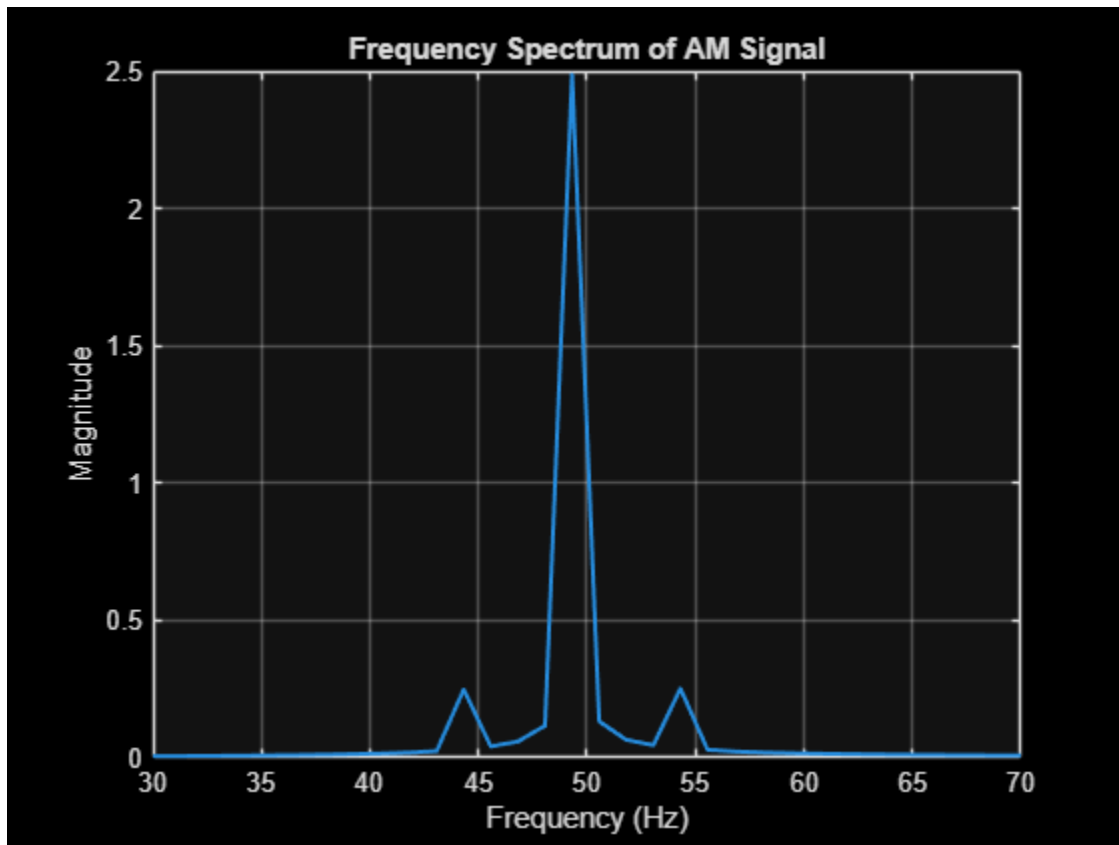
plot(f, AM_FFT, 'LineWidth', 1.5);
xlim([30 70]);
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Frequency Spectrum of AM Signal');
grid on;

% Display results
fprintf('Upper Sideband Frequency (USB) = %d Hz\n', USB);
fprintf('Lower Sideband Frequency (LSB) = %d Hz\n', LSB);

Upper Sideband Frequency (USB) = 55 Hz
Lower Sideband Frequency (LSB) = 45 Hz

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